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Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 21:55 UTC+8 Beijing Time)

Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.290 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 102.99 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The accelerated path executes embeddings, decoder layers, final normalization and tied-output GEMV without delegating accelerated tokens to the parent. Wide memory transport and independently owned weight buffers reduce transfer overhead; graph-specific fusion avoids architectural scratch writebacks. Parallel arithmetic increases storage and multiplier resources. A separate literal parent instance preserves legacy modes. The diagram represents interacting responsibilities, not a strictly serial execution sequence.

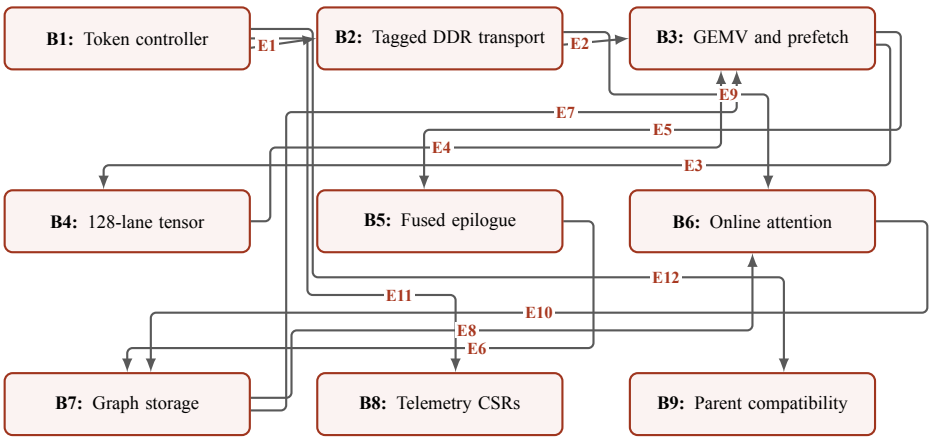


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token controller	Sequences embedding, attention and MLP phases; validates token position/phase; emits ordered logits and manages cancellation.
B2: Tagged DDR transport	Accepts logical 512-bit transactions, retains outstanding ownership and associates reordered responses with descriptors.
B3: GEMV and prefetch	Fetches payloads and scales, assembles two independently owned weight tiles and accumulates row partials in column order.
B4: 128-lane tensor	Uses exact group exponent bases and lane residuals, independently rounded products and seven balanced FP32 reduction levels.
B5: Fused epilogue	Provides RMSNorm replay, two-stage inverse square root, split-half RoPE, SwiGLU, residual addition and pass-through.
B6: Online attention	Broadcasts each causal KV vector across six distinct query states and retains running maxima, denominators and output accumulators.
B7: Graph storage	Holds residuals, BF16 operands, Q/K/V and necessary gate/RoPE branch joins using bounded, phase-owned storage.
B8: Telemetry CSRs	Counts actual activity, stalls, transfers, reuse and completion events; supports atomic snapshots and saturating counters.
B9: Parent compatibility	Instantiates unchanged anvil_q2_fp_top for legal legacy BF16/FP16, AWQ and int8-cache configurations.

Directed relationship	Meaning
E1: B1 to B2	Memory descriptors and cancellation control
E2: B2 to B3	Returned weights and scale metadata
E3: B3 to B4	Tagged weight and activation tiles
E4: B4 to B3	FP32 tile partial sums
E5: B3 to B5	Completed GEMV rows through controller
E6: B5 to B7	Normalized, rotated and joined values
E7: B7 to B3	Registered activation tiles
E8: B7 to B6	Six-head query command
E9: B2 to B6	Controller-assembled causal KV stream
E10: B6 to B7	Normalized attention outputs
E11: B1 to B8	Observed activity and completion events
E12: B1 to B9	Elaboration-selected alternative token path

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, cancel	Input	1 bit each	Single synchronous interface domain; asynchronous active-low transaction reset. Cancellation aborts local work but preserves accepted memory drain obligations.
Token and logit streams	Input/output	32-bit token_ id, token_pos, logit_index and logit_ data; 1-bit controls	Valid/ready handshakes; new_sequence restarts at position zero; token_phase selects prefill/decode. Logits are binary32 in vocabulary-index order. done follows final logit acceptance.
Parent mem_req_* / mem_rsp_*	Output requests/input responses	SB banks; 32-bit data/ coordinates, 64-bit tags	Preserves parent ports and meanings. Kind is 4 bits and projection is 3 bits per bank. Inactive memory interface request outputs are zero.
ddr_req_*	Output, with input ready	512-bit data; 64-bit byte strobes/tag; 32-bit layer/ row/column; 4-bit kind; 3-bit projection; 2-bit format	Logical controller boundary, not a DDR PHY. Every accepted read or write transfers one 64-byte beat and requires exactly one tagged response.
ddr_rsp_*	Input, with output ready	512-bit data, 64-bit tag, valid/error	Responses may reorder. At least two reads can remain outstanding, including QUEUE_DEPTH=1. Live tags cannot be reused; unknown or duplicate tags are faults.
CSR request/response	Input requests/ output responses	16-bit byte address, 32-bit data, 4-bit write strobes	One outstanding request; two registered response stages. Misaligned, unsupported and read-only writes return error. Accessible while token execution stalls.
Tensor streams	Internal input/ output	512-bit weight beats; 2048-bit activations; 128-bit mask; 32-bit tags/ results	Four BF16 or two MX beats assemble a 128-element tile. Each MX beat carries two scale bytes through w_ scale. Weight and activation tiles pair in order with equal tags.
Epilogue stream	Internal input/ output	32-bit operands, coefficients, index, length and tag; 3-bit mode	Modes 0–4 are RMSNorm, RoPE pair, SwiGLU, residual and pass-through. Other modes fail. RMS vectors cannot interleave; stalled payload and metadata remain stable.

Signal / group	Direction	Width	Contract
Attention streams	Internal input/ output	6*HEAD_ DIM BF16 queries; HEAD_DIM BF16 KV vectors; six FP32 output vectors	MX uses the low HEAD_DIM payload bytes and ceil(HEAD_DIM/32) scales. Command precedes KV items; indices must enumerate exactly 0 through cmd_ pos with cmd_len=cmd_pos+1.

Parameter	Default	Meaning / supported values
NUM_LAYERS	28	Accelerated range 1–28.
HIDDEN_SIZE / INTERMEDIATE_SIZE	1536 / 8960	Positive, at most defaults; multiples of eight; inherited divisibility/address checks apply.
NUM_Q_HEADS / NUM_KV_HEADS / HEAD_DIM	12 / 2 / 128	Q heads=6*KV heads; hidden=Q heads*HEAD_DIM; even HEAD_DIM=2–128; KV-head element product divisible by eight.
VOCAB_SIZE / MAX_ SEQ_LEN	151936 / 32768	Accelerated ranges 1–151936 and 1–32768.
GROUP_SIZE / AWQ_ BITS	128 / 4	GROUP_SIZE divides hidden/intermediate; AWQ_BITS= 4. MX grouping is independently fixed at 32.
BANKS / SB	4 / 4	BANKS=1,2,4; SB equals the safe derived bank width.
CACHE_BLOCK / QUEUE_DEPTH	16 / 3	CACHE_BLOCK is a power of two up to 65536 dividing HEAD_DIM; QUEUE_DEPTH=1–16.
FP_FORMAT / WEIGHT_MODE / CACHE_MODE	0 / 0 / 0	Accelerated BF16 uses format 0 and weight/cache modes 0=native or 2=MXFP8. Legal legacy mode-1 selections retain parent semantics.
ACCEL_ENABLE / COUNTER_BITS	1 / 64	ACCEL_ENABLE=0 selects parent; mode 2 requires acceleration and cannot mix with legacy modes. Counter width=1–64.
FlashAttention TILE_ TOKENS	8	Block-level range 1–32, including non-powers of two.

III Behavior and corner cases

A Arithmetic and compression

The numerical contract is binary32 round-to-nearest-even with gradual underflow, signed-zero distinctions and canonical NaN 0x7fc00000. Architectural nonfinite inputs/results fail. BF16 expansion is exact; narrowing is RNE. Group alignment retains full significands and exponent residuals rather than truncating to a common mantissa. MX uses E4M3FN payloads and E8M0 power-of-two scales; scale 255 and NaN payloads are invalid. Encoding selects the smallest scale accommodating the group maximum at payload magnitude 448; zero groups use scale 127.

B Reduction and fusion

Products round independently before the fixed adjacent-pair tree. GEMV tile sums accumulate in ascending column order. RMS squares and sums in index order, multiplies by rounded 1/N, adds epsilon 0x358637bd and applies inverse-square-root normalization. RoPE uses split-half pairs and theta=1000000. SwiGLU evaluates a stable sign-dependent sigmoid using exp(-abs(gate)); exponential arguments below -80 may produce positive zero. Only graph-required residual, RMS replay and branch-join values are retained.

C Pipeline timing

With acceptance at E0, an L-stage pipeline produces valid after E(L-1). Multiply/add/tensor latencies are 3/4/38 registered stages; inverse square root is two stages with initiation interval one. RoPE/residual/pass-through/SwiGLU use 7/4/1/24 stages. RMS inputs are spaced four clocks; its first output is valid 21 clocks after final input acceptance, without stalls.

D Attention

Each KV vector is consumed once and broadcast internally to six independent online-softmax states. Running-maximum changes rescale prior denominators and weighted sums. Final normalization follows the last causal item. Storage does not grow as a sequence-length score matrix. Duplicate indices, malformed last markers and noncausal commands fail.

E Reset, stalls and cancellation

Transfers use pre-edge valid&&ready. Public valid, ready, completion, write and payload outputs are zero during reset. Unreset memories and coefficient captures are protected by valid/ownership state and write-before-read. Stalled items retain complete payloads. Cancel wins over new token acceptance, suppresses new work and waits for accepted tagged requests to drain before cancel_done.

F Telemetry

Version 0x00020200 is at 0x0000; control at 0x0008 provides write-one clear and snapshot. Twelve 64-bit counter pairs span 0x0010–0x0068 for activity, stalls, underflow, bubbles, KV bytes, DDR beats, reuse and completions. Counters saturate and zero-extend. Clear dominates increments; simultaneous clear/snapshot captures zero. Recommendations do not change numerical modes.

IV Verification methods and measured results

A Behavior and compatibility

Method. Compact Icarus checks plus compiled Verilator matrices and independent scoreboards.

Result. Recorded suite passed: 12 legal startup configurations, 29 explicit parameter rejections, seeds 17/89/203, native/MX combinations, legacy compatibility, cancellation/reordering, compact 28-layer sequencing and full-sized operator exercises. This execution predates the nonlinear repair.

Evidence: E087

B Software-vs-Hardware numerical comparison

Method. Simulated RTL outputs compared with independent rational IEEE arithmetic and high-precision mathematics.

Result. Historical snapshot passed 65,536 BF16 patterns, 65,536 MX code/scale pairs, 10,134 finite balanced-tree comparisons, 27,940 positive rsqrt checks within four ULP, 1,000 RMS vectors, 10,000 RoPE pairs, 10,000 SwiGLU elements and 100 six-head attention transactions. Current nonlinear arithmetic has no post-repair numerical execution; no physical FPGA numerical comparison is supplied.

Evidence: E088, E1010, E042

C Defect sensitivity

Method. Disposable, successfully compiled source mutations exercised through genuine comparisons.

Result. All ten defects detected: lane omission, exponent alignment, truncation, buffer association, rsqrt coefficient, RoPE mapping, sigmoid sign, softmax rescaling, sixth-head mapping and telemetry stall omission. Results apply to the recorded snapshot.

Evidence: E089

D Five-domain measured performance

Method. Matched compiled parent/FP2 simulations; seeds 17/89/203; native and MX; 2−17-cycle memory latency and reordered responses.

Result. Thirty parent/FP2 comparisons produced 36.70165x geometric-mean speedup. Seed-17 parent/native/MX cycles were language 1,018,340/25,261/30,805; mathematics 676,208/16,589/20,258; coding 1,018,340/25,261/30,805; reasoning 1,352,037/33,020/40,537; knowledge 1,018,340/25,261/30,805. All 30 cases improved over parent, but compact MX was slower than native. The complete seed/mode traffic, precision, telemetry and hash ledger is evidence/fp2-benchmark-20261102T231600Z-b59f0780/summary.json.

Evidence: E091

E Bandwidth, broadcast and fusion

Method. Accepted-beat accounting, executed nonbroadcast reference and public prefetch ownership checks.

Result. Payload density was exactly 2x; three-row lengths 1536/8960 achieved metadata-inclusive density gains 1.920x/1.931x and native-to-MX cycles 706→611/3934→3186. All 30 broadcast comparisons reduced KV reads by 83.333%. Intermediate architectural writes were zero. A directed prefetch test demonstrated simultaneous fill/consume and both buffers owned; long-stream runs themselves recorded no overlap.

Evidence: E091

F Generic synthesis

Method. Yosys lowering, optimization, hierarchy/state checks and zero-latch/divider assertions.

Result. Recorded default, named-block, mode and parent cases passed. Comparable compact native parent/FP2 state was 17,605/169,913 bits; inferred memory 208/1,859,200 bits; generic multiplier cells 34/286. These are pre-repair logical proxies, not physical area or power.

Evidence: E090

G L1 RTL timing lint

Method. Native source-bound timing-hazard inspection.

Result. Passed across 15 current RTL files: no combinational divider/modulo, mixed-assignment, LUT-clock-gating or AST-unroll hazards.

Evidence: E1008

H L2 Quartus TimeQuest STA

Method. Intel Quartus Prime 17.1 post-fit STA at 10.000 ns.

Result. Passed: WNS_setup = +0.290 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 102.99 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E1009

I L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical PCIe bridge register probes.

Result. All three passed deterministically. Receipt tier is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; it establishes bridge connectivity, not accelerator bitstream deployment, inference-vector execution or software/FPGA numerical parity.

Evidence: E1010

J Portable delivery

Method. Fresh copy of 67 declared files; normal and isolated Python launches.

Result. Both copied launches passed compact simulation, independent comparisons and synthesis reproduction. The copied nonlinear source differs from the current repaired file, so current-source portability is not established by this historical result.

Evidence: E092

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 21	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

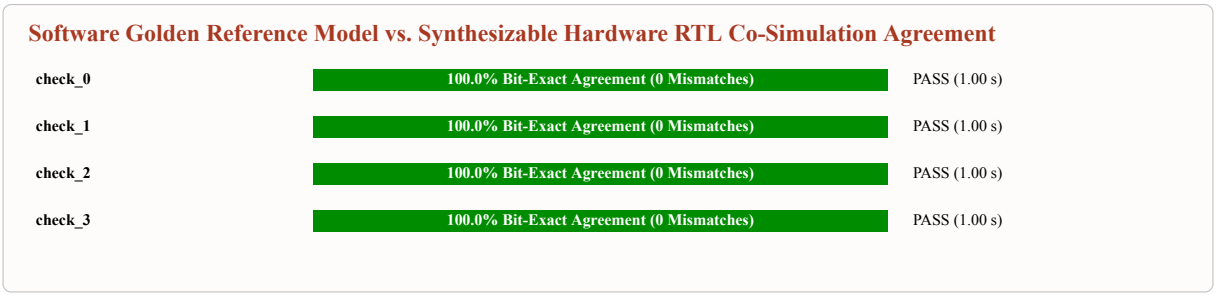


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.29 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.